Trait anxiety modulates the electrophysiological indices of rapid spatial orienting towards angry faces

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We investigated the electrophysiological markers of attentional bias for threat in anxiety. Low-anxiety and high-anxiety individuals performed a spatial-cueing task, in which an emotional facial expression (angry or happy) was presented alongside a neutral expression. Results revealed that angry expressions elicited an enhanced N2pc component, but that this was true only for those reporting high levels of trait anxiety. These results confirm the early capture of spatial attention by threat-related stimuli, and demonstrate that this early bias is modulated by trait anxiety. Enhanced P1 amplitudes to targets after presentations of angry expressions were also found; however, this effect was not modulated by trait anxiety levels. Our findings indicate that individual differences in temperament are an important determinant of the early neural response to threat. *NeuroReport* 19:259–263 © 2008 Wolters Kluwer Health | Lippincott Williams & Wilkins.

Keywords: angry facial expressions, emotion, N2pc, selective attentional bias, spatial orienting, trait anxiety

Introduction

Evolutionary pressures dictate that potential dangers in the environment should be rapidly detected. Both behavioural and neuroimaging researches confirm that attentional mechanisms do indeed prioritize threat-related stimuli, and that neural circuits centred on the amygdala are crucial for the efficient processing of threats [1]. It is becoming clear that individual differences in temperament, especially those relating to neuroticism or trait anxiety, are an important determinant of the degree to which threatening stimuli can capture attention [2]. Studies using functional magnetic resonance imaging have also demonstrated enhanced amygdala reactivity to threat in highly trait-anxious individuals [3].

A problem with all foregoing research, however, is that they do not shed light on the temporal dynamics of attentional processing. The use of event-related potentials (ERPs), derived from analyses of scalp electrophysiology, is ideally suited to investigate the early neural response to threat. The N2pc, for example, is one ERP component that has been shown to provide an uncontaminated index of attentional selection [4,5]. The N2pc is a negative-indicating voltage deflection that occurs at posterior electrodes, contralateral to the position of a target: thus, its isolation from the rest of the ERP waveforms in bilateral displays is possible. The N2pc consists of an early response (170–220 ms after stimulus onset) originating in the parietal cortex, and a later response (225–270 ms after stimulus onset) originating in the occipitotemporal regions [6]. Recently, it has been demonstrated that task-irrelevant fearful faces elicited an N2pc, providing direct electrophysiological evidence that threat-related stimuli could bias the distribution of spatial attention [7]. Whether or not this enhanced N2pc response to threat is modulated by individual differences in trait anxiety is unknown.

The main aim of this study was to determine whether the N2pc response to threat was modulated by trait anxiety. A secondary aim was to assess whether the sensory processing of targets could be enhanced by presenting them in a location that had been occupied by an angry expression until then. In a covert-orienting task [8], validly cued targets induced a larger P1 component (originating in the extrastriate visual cortex) compared with invalidly cued targets [9–12]. This P1-to-target component can be further enhanced by fearful faces [13] and by pictures depicting negative affective scenes [14]. This study aimed to extend these findings, by determining whether targets following angry facial expressions benefitted from enhanced sensory processing, and whether trait anxiety modulated this effect.

To summarize, the role of trait anxiety in determining the electrophysiological response to the initial onset of a threat-related stimulus remains unknown. We thus examined whether trait anxiety modulated the N2pc component to angry facial expressions. We predicted that an enhanced N2pc to the onset of an angry facial expression would be enhanced by high levels of trait anxiety.

Method

Participants

Twenty-eight volunteers (19 women) were selected from a larger pool of University of London students who had completed the trait-anxiety scale of the State-Trait Anxiety Inventory (STAI: see Ref. [15]). High-anxiety (HA) individuals (N=14) were defined as those scoring 45 or more on the STAI trait-anxiety scale and the low-anxiety (LA) individuals (N=14) were defined as those scoring 35 and below on this scale. Mean trait anxiety for the HA individuals was 50.8 (SD=4.8); for the LA individuals, it was 30.0 (SD=5.2). The mean age for the HA group was 31.9 years (SD=5.0), and for the LA group, 30.6 (SD=7.2). All participants had self-reported normal or corrected-to-normal eyesight.

Stimuli and task

The pictures of eight individuals were selected from the Karolinska Directed Emotional Faces set [16]: four (two male faces) with happy expressions and four (two male faces) with angry expressions. Each emotional face was paired with its corresponding neutral face. The faces were trimmed of all nonfacial features, enclosed in a rectangular frame subtending to $5.8 \times 8.7^{\circ}$ (7.1×10.6 cm) of the visual angle at a 70-cm viewing distance, and placed on a black background. All faces were converted to an 8-bit greyscale, and were then equated for mean pixel luminance, contrast, and centre-spatial frequency using Matlab [17]. The face-pair stimuli were presented bilaterally 7.3° (9 cm) along the horizontal meridian from the central-fixation cross. The central-fixation cross subtended to a $1.6 \times 1.6^{\circ}$ visual angle (2×2 cm) and to 0.08° (0.1 cm) in thickness. The target stimulus was a white horizontal or vertical bar subtending to $4.6 \times 0.32^{\circ}$ (5.6×0.4 cm), which was presented laterally at 7.3° along the horizontal meridian from the central fixation. The task was presented on a 19-inch Dell TFT (Bracknell, Berkshire, UK) screen through an Intel Pentium 4 computer, running DMDX (Tucson, Arizona, USA) [18] presentation software.

Each trial began with a fixation cross being presented at the centre of the screen for 500 ms (this cross remained on the screen until the offset of the target, and participants were asked to fixate it throughout the experiment). After this, either an angry–neutral or happy–neutral face pair appeared for 150 ms (Fig. 1). The emotional face appeared as many times on the left side as on the right. This presentation was followed by a blank screen for either 150 ms (short cue–target onset asynchrony (CTOA)) or 600 ms (long CTOA). This was succeeded by the target that was shown for 150 ms, followed by a blank screen for 1650 ms. At the onset of target, either the horizontal or vertical portion of the cross became thicker (0.32°, 0.4 cm). Participants were instructed to press the appropriate button on the response box using the index finger of their dominant hand only when the orientation of the target matched that of the thicker portion of the cross. They were instructed to ignore the faces, but to make a judgement about the target while fixating the cross.

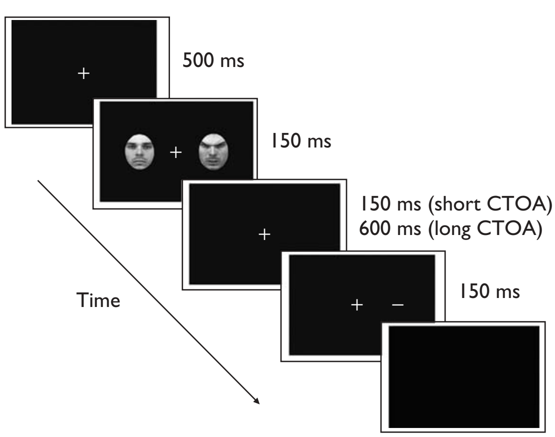

Fig. 1 An example of a no-go trial with the angry–neutral face stimuli.

The experiment was divided into 16 blocks of 96 trials. Each block included (i) 64 'no-go' trials (two emotional cues: angry and happy; two emotional cue locations: left and right; two target locations: valid and invalid; two CTOAs: short and long; and four repetitions, resulting in $2 \times 2 \times 2 \times 2 \times 4$=64 trials), in which the orientations of the target and the cross did not match; (ii) 16 'go' trials (two emotional cues: angry and happy; two emotional cue locations: left and right; two target locations: valid and invalid; and two CTOAs: short and long, giving $2 \times 2 \times 2 \times 2$=16 trials), in which the orientations of the target and the cross did match; and (c) 16 'catch' trials (two emotional cues: angry and happy; two emotional cue locations: left and right; and four repetitions, giving $2 \times 2 \times 4$=16 trials), in which no target was presented and the fixation cross remained on the screen for 1400 ms after the offset of cue. Participants were instructed to only respond on the go trials. In half of the trials, the target orientation was vertical, and in the other half, horizontal. All trials within a block were randomly presented. Speed and accuracy were emphasized.

Procedure

Participants were told that they would be doing a task on visual perception. They completed the relevant consent and information forms about the experiment. The electrode cap was fitted and impedances were kept below 5 kΩ.

The experiment took place in a dimly lit, sound-attenuated cabin. Instructions were read by the participant, and were further emphasized by the experimenter. A separate practice block of 12 neutral face-pair trials (eight no-go, two go, and two catch trials) was completed. Participants were thanked, paid, and debriefed at the end of the session.

Electroencephalogram recording

Electroencephalogram (EEG) was recorded (Brain Products GmbH, Gilching, Germany) from 26 Ag–AgCl scalp electrodes mounted in a lycra electrode cap (Easycap GmbH, Herrsching-Brietbrunn, Germany) according to the international 10–20 system. Horizontal eye movements (horizontal electrooculogram; HEOG) were monitored by placing electrodes laterally on the outer canthi of the eyes. HEOG was then derived by calculating the difference between the two HEOG electrodes. All electrode sites were referenced to FCz and then re-referenced offline to the average of all electrodes. The EEG was amplified to between 0.1–125 Hz and sampled at 500 Hz. The EEG was then band-pass filtered offline with cutoff frequencies of 1–30 Hz.

For face stimuli, the EEG was epoched offline into 400-ms windows, including a 100-ms prestimulus baseline. For target stimuli, the EEG was epoched into 300-ms windows from the onset of the target. Epochs containing amplitudes exceeding $\pm 40\,\mu V$ in the Fpz electrode or in the HEOG electrode were excluded. Trials with amplitudes exceeding

$\pm 80\,\mu V$ were discarded, as were trials containing incorrect responses.

Data analysis

Analysis of ERPs was confined to the occipitoparietal electrodes, PO7 and PO8, where the visual components are most prominent [9]. ERP components were quantified as the mean amplitude within a specified time window with respect to the mean amplitude of the baseline interval (100 ms preceding face onset).

The N2pc was quantified by subtracting the signal recorded from the ipsilateral electrodes (with respect to the visual field of the emotional face) from that of the contralateral electrode. The epoched waveforms corresponding to the emotional faces in the left and right visual fields were collapsed to eliminate any hemispheric asymmetries that might be unrelated to attention [12]. The N2pc was divided into two successive time windows: early N2pc (170–220 ms after stimulus onset) and late N2pc (225–270 ms after stimulus onset). Both N2pc windows were maximal at the posterior electrodes sites, PO7/PO8; therefore, the analysis concentrated on these sites. The N2pc yielded the following within-participant variables: emotional expression (two) and contralaterality (two).

For ERPs elicited by the target onset (bar probe), the P1 component was analysed within the time window of 100–150 ms from target onset and was maximal over the occipitoparietal electrodes, PO7/8. The P1 for the trials in which the target replaced the emotional face (valid trials) were compared with when it replaced the neutral face (invalid trials) for both CTOA durations. This yielded the following within-participant variables: emotional expression (two), CTOA (two), validity (two), visual field of target (two), and hemisphere (two).

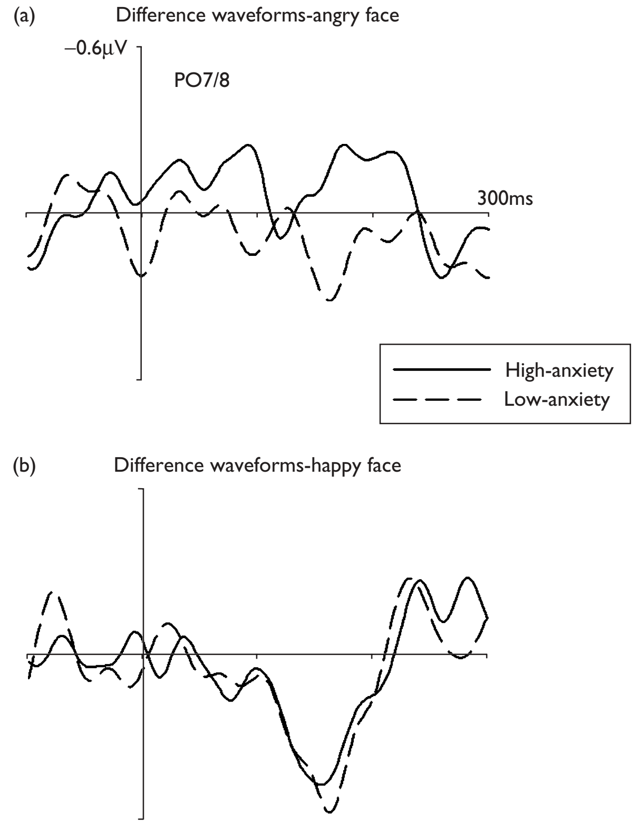

Fig. 2 Difference waveforms obtained by subtracting event-related potentials (ERPs) at ipsilateral electrodes to the side of emotional face from ERPs at contralateral electrodes in angry−neutral (a) and happy−neutral (b) trials for high-anxiety (HA; solid lines) and low-anxiety (LA; dashed lines) individuals.

Results

Effects of trait anxiety on the N2pc component to threatening faces

ERPs were measured contralateral and ipsilateral to the emotional face on no-go and catch trials averaged across the left and right visual fields of the emotional face. An enhanced negativity contralateral to the angry face was observed for HA individuals in the early window of 170–220 ms after stimulus onset. This effect is clearly seen in Fig. 2, in which the difference (contralateral−ipsilateral) waveforms are depicted.

A $2 \times 2 \times 2$ mixed analysis of variance (ANOVA) with 'emotional expression' of face (angry and happy) and 'contralaterality' (electrode being placed contralateral vs. ipsilateral to the side of the emotional face) as within-participant factors and 'group' (HA and LA) as a between-participant factor revealed a marginal main effect of emotional expression [$F(1,26)=3.11$, $P=0.08$], a two-way interaction of emotional expression × contralaterality [$F(1,26)=9.76$, $P<0.005$], and a three-way interaction of emotional expression × contralaterality × group [$F(1,26)=4.20$ $P=0.05$].

To explore the three-way interaction, separate ANOVAs were conducted for each of the anxiety groups with emotional expression and contralaterality as within-participant factors. For the LA, the two-way interaction of emotional expression and contralaterality was not significant [$F(1,13)<1$], but for the HA it was highly significant [$F(1,13)=13.65$, $P<0.004$]. The mean difference between contralateral and ipsilateral electrodes within the early N2pc window was subjected to a series of paired sample t-tests for each of the LA and HA groups with a Bonferroni-corrected P value of 0.0125. The HA group showed a significantly enhanced N2pc for angry faces [$t(13)=3.09$, $P<0.01$], whereas the LA group did not [$t(13)=1.14$, $P=0.27$]. Neither group showed an N2pc for happy faces. Analyses in the late window revealed no significant effects for either group (all ts<2.25).

Effects of threatening cues on the P1 component to target stimuli

The P1 amplitudes to the onset of target were analysed and a higher-order interaction involving 'visual field' and 'hemisphere' was found. Therefore, separate analyses were conducted for electrodes contralateral and ipsilateral to the visual field of target. The contralateral P1 component was analysed by a $2 \times 2 \times 2 \times 2$ mixed ANOVA (emotional expression: angry and happy; validity: valid and invalid; CTOA: short and long; visual field of target: left and right; and anxiety group: HA and LA). A significant emotional expression × validity interaction [$F(1,26)=5.44$, $P=0.02$] and a three-way interaction of emotional expression × validity × CTOA [$F(1,26)=5.46$, $P=0.02$] were found (Fig. 3). The P1 amplitude was greater for valid, compared with invalid, targets that were preceded by angry faces at the short CTOA

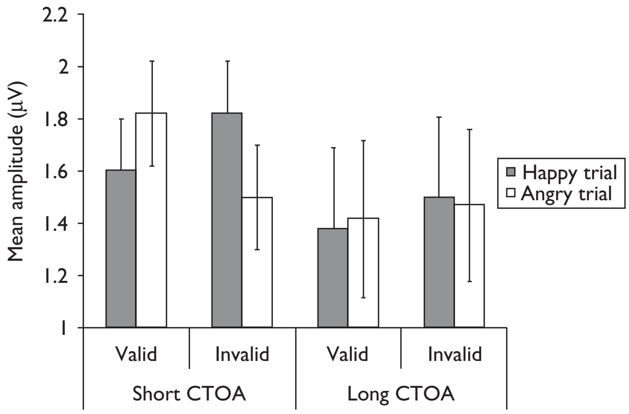

Fig. 3 Mean P1 amplitudes to validly and invalidly cued angry and happy targets at short and long cue–target onset asynchronies (CTOAs).

[1.81 vs. 1.56 μV; $t(27)=2.45$; $P=0.02$]. The trend for the P1 amplitudes of invalidly cued happy targets to be greater than validly cued happy targets was nonsignificant [1.82 vs. 1.60 μV; $t(27)=1.9$; $P=0.06$]. No significant effects were observed at long CTOA. No significant interactions were observed with group. For ipsilateral components, the analysis observed no significant interactions with validity or group.

Discussion

We show for the first time that the early N2pc component of attention can be elicited for angry facial expressions of emotion. Crucially, we also found that an individual's level of trait anxiety was an important determinant of this early electrophysiological response to a threatening stimulus. HA individuals showed rapid exogenous orienting of spatial attention to threatening cues, as indicated by the N2pc component of attention. This early attentional capture by threat in anxiety might be a neural marker for the development of the attentional biases observed behaviourally, which are a known risk factor for anxiety disorders [2,19]. The temporal dynamics of this early bias supports the theoretical view that anxiety is associated with an enhanced early shift of attentional resources towards threat [2,20], which also concurs with evidence from functional magnetic resonance imaging that linked self-reported anxiety to heightened amygdala activity for unattended fearful expressions [21,22].

Our findings are consistent with recent theoretical developments linking the adverse effects of anxiety to an imbalance between stimulus-driven bottom-up processes and top-down goal-directed processes in working memory [23]. Accordingly, anxiety increases the influence of the stimulus-driven processes over top-down attention-control mechanisms [23]; this might be due to the combined effects of a heightened amygdala response to threat alongside a reduced recruitment of the prefrontal cortical areas of the brain, which are thought to modulate the regulation of threat [3]. Our results suggest that this early activity within the parietal cortex, as indexed by an early N2pc, enhances the stimulus-driven aspects of a threat response, leading to greater distractibility towards threat in HA people [23,24].

Our second aim was to examine whether the P1 amplitude would also be enhanced for targets following angry facial expressions and their modulation by reported trait anxiety. Enhanced P1 amplitudes replacing fearful facial expressions were previously demonstrated [13]. We found enhanced P1 amplitudes to targets replacing angry facial expressions and not neutral facial expressions. This effect, however, only occurred at the short CTOA (300 ms) and not at the longer CTOA of 750 ms. This finding is important in showing that the presence of an angry expression leads to a relatively brief enhancement of the sensory processing of a target presented in the same spatial location, and further supports the notion that the presence of threat induces a rapid covert orienting of spatial attention to its location [1].

The observed P1 effect was not modulated by the degree of trait anxiety, in contrast to other researches indicating such an effect [14]. It is likely that methodological differences account for the contrasting results between that study and the current one. In the former study [14], a single stimulus was presented for a relatively long time (600 ms). In our study, however, a bilateral display was presented for just 150 ms. Given the division of attention between the two items in this study, along with the short presentation time, this study provides a measure of the covert orienting of attention. In contrast, the former study [14] is more likely to assess the later processes involved in the maintenance of attention to a particular stimulus. The anxiety-related differences found therein might be a reflection of an anxiety-related delay in disengaging attention from threat, as has been found behaviourally [24].

Conclusions

Angry faces elicited an enhanced N2pc, as well as an enhanced P1 response, to subsequent targets. The N2pc effect to angry faces was, however, strongly modulated by the degree of self-reported trait anxiety. This is consistent with the growing evidence that differences in some of the other 'big five' personality traits are also important determinants of the neural response to different classes of stimuli [25].

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